

Next up on the list comes a topic that many analysts spend a great deal of time deliberating. Feature engineering or column generation seems like a job for the experts, but the right research and tools make it possible for anyone to do it. In order to generate a significantly accurate model that is adaptable and useful, it is never a good idea to stick with the columns that were obtained originally. In simple terms, depending on the kind of data that the user may be dealing with, there should be more columns and values generated. Take the example of a simple air quality predictor that measures atmospheric pollution. Sensors would pick values for temperature, pressure, concentration, air mixing strength and wind speed. Instead of just dumping these variables straight to the model, generate new insights from it. One can easily calculate values for dewpoint, time taken, cloud temperature, solar intensity and cloud height using some simple calculations.

Feature engineering can be conducted well in advance after the data has been collected or after the cleaning step. Just be certain that not all features would be useful to the dataset as a whole. For example, if the new generated

Data Preparation Processes



Data Ingestion & Profiling

Prepare, search, sample, profile, catalog, inventory data assets, as well as tag & annotate data.



Data Quality & Security

Data governance, stewardship, data encryption, user permissions & data lineage, data masking, authentication, LDAP/SSO



Data Catalog & Metadata Management

Create, catalog data sources, data sources, metadata, cataloging of data sources, transformations, user activity; source attributes, lineage/relationship & APIs



Data Enrichment

Entity extraction, attribute development based on domain, use machine learning to enhance datasets



Data Structuring & Modeling

Data modeling & structuring that allows users to specify data types & relationships



Data Transformation

Data mashup, blending, cleansing; filtering; user-defined calculations, groups & hierarchy sets



Operationalization

Meet business needs with data sources & data types for business success

Data Preparation Steps